Gayathri Prakash Menakath

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Summary

Full Stack Software Engineer with 4+ years of experience designing and building scalable, data-intensive applications using TypeScript, React and Python. Experienced across the full development lifecycle, spanning frontend applications, backend services, APIs, database architecture and cloud infrastructure. Strong expertise in geospatial platforms, distributed workflows and data processing, with a focus on building reliable, maintainable systems and delivering complex technical solutions.

Experience

Full Stack TypeScript Engineer

AgriProve, Australia *June 2025 – Present*

- Developed features for a geospatial platform using React, TypeScript and GraphQL, transforming complex spatial data into interactive map visualisations and responsive user interfaces.
- Architected and built a high-performance reporting engine using HTML and Puppeteer, generating 12-page, print-quality PDFs with 300 DPI maps. Integrated Claude-assisted content generation and configurable map layouts.
- Own the development of the company's carbon-crediting calculator, implementing the legislated Australian soil carbon method (CER Schedule 2) in Python, with tested calculations reconciled against lodged offset reports.
- Designed and built an automated carbon project setup pipeline using PostGIS, Temporal and Python, replacing manual QGIS workflows with automated spatial analysis, constraint detection, sample-point generation and shapefile processing.
- Led the migration of background processing from AWS SQS to PostgreSQL using pg-boss, simplifying system architecture and improving reliability. Developed asynchronous HubSpot CRM integrations and GraphQL APIs.
- Improved application observability and troubleshooting using Honeycomb and FullStory, leveraging Claude Code for development, testing and technical planning.
- **Technologies:** React, TypeScript, JavaScript, GraphQL, Apollo, Python, PostgreSQL, PostGIS, Temporal, pg-boss, Puppeteer, Honeycomb, FullStory, Claude, Ansible, GitHub Actions.

Software Engineer

Voiseed, Milan, Italy *Jan 2023 – Nov 2024*

- Developed and maintained a cloud-based web application ensuring high performance and a seamless user experience, while integrating GraphQL to enable smooth communication between the frontend and backend services.
- Implemented JavaScript AWS Lambda functions using Amplify to interact with DynamoDB, enabling efficient data retrieval and storage. Integrated these functions with pre-deployed deep learning models on AWS, facilitating seamless inference calls.
- Managed deployments and CI/CD workflows using AWS Amplify, ensuring smooth and automated application updates.

- Configured Grafana and Prometheus dashboards to monitor system health and improve alerting accuracy.
- Maintained a Docusaurus-based documentation project, providing comprehensive technical documentation for the web application.
- **Technologies:** ReactJS, Redux, TypeScript, Material UI, AWS (Lambda, DynamoDB), GraphQL, Prometheus, Grafana.

Intern
Voiseed, Milan, Italy *Aug 2021 – Dec 2022*

- Developed fast and scalable websites using Gatsby's static site generator, ensuring efficient data retrieval and rendering.
- Integrated Strapi as a flexible content management system, enabling dynamic content handling, role-based access control, and streamlined API management.
- **Technologies:** ReactJS, GraphQL, Git, Strapi, Gatsby, AWS EC2, Fargate, RDS.

Software Developer
Asteria Aerospace,Bangalore, India *2018 – 2019*

- Contributed to building software for uploading data from drones to AWS, retrieving it for analysis, and processing aerial imagery such as orthophotos, DSM, and DTM tiles.
- Integrated Auth0 for user authentication, leveraging JSON Web Tokens for secure access.
- Worked with Go authorization libraries like Casbin to implement role-based access control (RBAC) and other access control models.
- Built and maintained APIs in Go, utilizing gRPC for communication and Protocol Buffers for data serialization.
- Automated the ingestion and grouping of AWS platform logs (CloudWatch) and application logs into a centralized logging service.
- **Technologies:** Golang, AWS, gRPC, Protocol Buffers, Docker, PostgreSQL, Git.

Projects

Unified Monitoring System
Master's Thesis, Politecnico di Milano *2022*

- Developed a centralized monitoring system for Voiseed, an AI-driven text-to-speech startup, integrating scalable data processing pipelines with AWS Kinesis, Athena, and Glue to monitor and process critical system data, improving operational efficiency and early issue detection.
- Focused monitoring efforts on four key areas: customer activities, deep learning model performance, AWS services, and local computing resources.
- Implemented operational monitoring for deep learning models using TorchServe, enabling proactive issue detection and timely intervention.
- Optimized AWS service monitoring to ensure reliability, cost efficiency, and performance.
- Deployed infrastructure monitoring for backend components and managed GPU resource tracking to optimize resource allocation and maximize data center efficiency.

- **Technologies used:** AWS - Kinesis, S3, Athena, Glue, Prometheus, Grafana.

Education

M.Sc. in Computer Science and Engineering, *Politecnico di Milano, Italy*
Key Coursework: Software Engineering, Business Information Systems, Unstructured and Streaming Data Engineering, Machine Learning, Natural Language Processing.

B.Tech in Computer Science and Engineering, *Amrita Vishwa Vidyapeetham, India*
CGPA: 8.58/10

Technical Skills

Programming Languages: TypeScript, JavaScript, Python

Frontend Development: React, Next.js, Redux, GraphQL, Material UI, Chakra UI, HTML, CSS

Backend Development: Node.js, AWS Lambda, Gin-Gonic, gRPC, REST APIs, Strapi

Databases: PostgreSQL, PostGIS, MySQL, DynamoDB

Cloud & DevOps: AWS (Lambda, Amplify, API Gateway, EC2, S3, CloudWatch, Athena, Kinesis, Glue, Fargate), Docker, Ansible, GitHub Actions

Geospatial & Workflows: Temporal, pg-boss, GeoJSON, Martin, STAC

Monitoring & Logging: Honeycomb, FullStory, Grafana, Prometheus, CloudWatch

Collaboration & Project Management: Git, Atlassian Suite, Figma, Agile (Scrum/Kanban), Claude Code